

Privacy-Preserving Federated Learning and Information-Theoretic Generalization Bounds

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Federated learning $\times$ Differential privacy $\times$ Generalization theory

Federated learning side

- Data stay local.
- Users exchange model updates, not raw samples.
- The released model can still leak information.

Information-theoretic side

- Privacy is distributional indistinguishability.
- Generalization is algorithmic stability.
- Mutual information and divergence connect both.

Core question: how do privacy noise, subsampling, and communication constraints control generalization?

Federated learning: the basic statistical model

Data model

There are n users. User i owns local data $Z_i \sim P_Z$. The learner outputs a model $W = A(S)$ from

$$S = (Z_1, \dots, Z_n), \quad W \in \mathbb{R}^d.$$

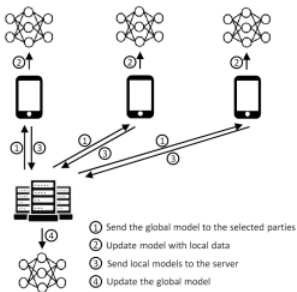
Federated constraint

The aggregator should not observe raw data. It only receives messages such as gradients, parameters, or noisy updates.

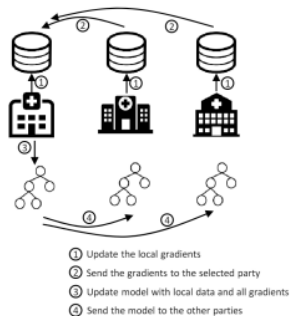
Statistical tension

Useful learning requires the model to depend on data; privacy asks that changing one user's data should barely change the output distribution.

Two federated communication patterns



(a) FedAvg



(b) SimFL

FedAvg: central server aggregates local models.

SimFL: decentralized update through selected parties.

Why privacy is not only cryptography

Secure aggregation protects messages

Encryption can hide individual updates during transmission, but the **final released model** may still encode information about a participant.

Differential privacy protects the output distribution

A randomized algorithm A is (ϵ, δ) -DP if for neighboring datasets S, S' and every event B ,

$$\mathbb{P}(A(S) \in B) \leq e^\epsilon \mathbb{P}(A(S') \in B) + \delta.$$

DP is therefore a statement about **how much one sample can move the law of the released model**.

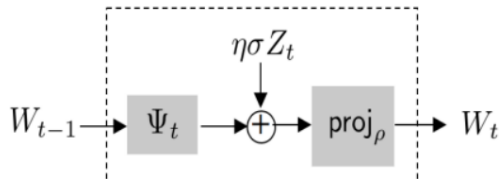
The projected noisy update as a Markov kernel

At round t , the update can be written as

$$\Psi_t(w) = w - \frac{\eta}{m} \sum_{i \in B_t} \nabla \ell(w, Z_i),$$

followed by Gaussian noise and projection:

$$K_t(w) = \text{proj}_\rho \left(\mathcal{N}(\Psi_t(w), \eta^2 \sigma^2 I) \right).$$



Interpretation

Training is a composition of Markov kernels:

$$\mu_T = \mu_0 K_1 K_2 \cdots K_T.$$

Privacy analysis becomes a question about how divergences contract through this noisy channel.

f -divergence and the DP divergence

f -divergence

For a convex function f with $f(1) = 0$,

$$D_f(\mu\|\nu) = \mathbb{E}_\nu \left[f \left(\frac{d\mu}{d\nu} \right) \right].$$

$\mathcal{E}_\varepsilon$ -divergence

DP can be expressed through the hockey-stick divergence

$$\mathcal{E}_\varepsilon(\mu\|\nu) = \sup_A [\mu(A) - e^\varepsilon \nu(A)].$$

Then A is (ε, δ) -DP iff

$$\mathcal{E}_\varepsilon(P_{A(S)}\|P_{A(S')}) \leq \delta$$

for all neighboring S, S' .

Closed form for Gaussian mechanisms

For two Gaussians with identical covariance,

$$N_1 = \mathcal{N}(m_1, \sigma^2 I), \quad N_2 = \mathcal{N}(m_2, \sigma^2 I),$$

define

$$k = \frac{\|m_2 - m_1\|_2}{\sigma}.$$

Then

$$\mathcal{E}_\varepsilon(N_1 \| N_2) = Q\left(\frac{\varepsilon}{k} - \frac{k}{2}\right) - e^\varepsilon Q\left(\frac{\varepsilon}{k} + \frac{k}{2}\right),$$

where

$$Q(t) = \int_t^\infty \frac{1}{\sqrt{2\pi}} e^{-u^2/2} du.$$

Key compressed notation

$$\theta_\varepsilon(r) = Q\left(\frac{\varepsilon}{r} - \frac{r}{2}\right) - e^\varepsilon Q\left(\frac{\varepsilon}{r} + \frac{r}{2}\right).$$

Contraction: the information-theoretic engine

A Markov kernel K contracts a divergence if applying K makes distributions closer:

$$\eta_f(K) = \sup_{\mu, \nu: D_f(\mu \| \nu) \neq 0} \frac{D_f(\mu K \| \nu K)}{D_f(\mu \| \nu)}.$$

For total variation, Dobrushin's coefficient has the two-point form

$$\eta_{\text{TV}}(K) = \sup_{y_1, y_2} \text{TV}(K(y_1), K(y_2)).$$

For DP divergence, the analogous quantity is

$$\eta_{\varepsilon}(K) = \sup_{y_1, y_2} \mathcal{E}_{\varepsilon}(K(y_1) \| K(y_2)).$$

Gaussian kernel on a bounded set Y

If $K(y) = \mathcal{N}(y, \sigma^2 I)$ and $\|Y\| = \max_{y_1, y_2 \in Y} \|y_1 - y_2\|_2$, then

$$\eta_{\varepsilon}(K) = \theta_{\varepsilon}\left(\frac{\|Y\|}{\sigma}\right).$$

Privacy composition through training rounds

Let μ_t and μ'_t be the laws of W_t under neighboring datasets S, S' . Data processing gives

$$\mathcal{E}_\varepsilon(\mu_T \| \mu'_T) \leq \mathcal{E}_\varepsilon(\mu_t \| \mu'_t) \prod_{j=t+1}^T \eta_\varepsilon(K_j).$$

Meaning

Only the round that touches the changed user can create privacy leakage. Later noisy rounds contract that leakage.



Subsampling and the final DP bound

Suppose $m = qn$ users are sampled without replacement per round, so $T = n/m = 1/q$. Assume the loss is convex, L -Lipschitz and smooth, and the update is projected to $\mathbb{B}_\rho$.

One clean way to read the theorem

The final privacy leakage is bounded by

$$\delta_\epsilon \leq q \theta_\epsilon \left(\frac{2L}{\sqrt{m}\sigma} \right) \sum_{s=0}^{T-1} \theta_\epsilon \left(\frac{2\rho\sqrt{m}}{\eta\sigma} \right)^s.$$

Equivalently,

$$\delta_\epsilon \leq q \theta_\epsilon \left(\frac{2L}{\sqrt{m}\sigma} \right) \frac{1 - a^T}{1 - a}, \quad a = \theta_\epsilon \left(\frac{2\rho\sqrt{m}}{\eta\sigma} \right).$$

Larger noise σ and smaller sampling rate q reduce privacy leakage; larger radius ρ or larger sensitivity L increase it.

Proof idea: where the pieces enter

- ① **Neighboring datasets:** S and S' differ in one user's data.
- ② **Creation of leakage:** if the changed user is selected, the update means differ by at most

$$\|\Psi_t(w) - \Psi'_t(w)\|_2 \leq \frac{2L\eta}{m}.$$

- ③ **Gaussian formula:** the immediate divergence is controlled by $\theta_\varepsilon(2L/(\sqrt{m}\sigma))$.
- ④ **Future contraction:** later kernels shrink the divergence by factors $\eta_\varepsilon(K_j)$.
- ⑤ **Subsampling:** the changed user is selected with probability $q = m/n$.

Information-theoretic viewpoint

The released model is the output of a noisy channel from data to parameters. DP follows from channel contraction plus sampling amplification.

Generalization error as a replace-one stability identity

For a learning algorithm A and sample S , define

$$\Delta_A(S) = L_{\mathcal{D}}(A(S)) - L_S(A(S)).$$

Let $S^{(i)} = (Z_1, \dots, Z'_i, \dots, Z_n)$ be S with the i -th point replaced by an independent copy. Then

$$\mathbb{E}[\Delta_A(S)] = \frac{1}{n} \sum_{i=1}^n \mathbb{E}[\ell(A(S), Z'_i) - \ell(A(S^{(i)}), Z'_i)].$$

Meaning

Generalization is controlled by how much the output model changes when one sample is replaced.

This is exactly the same stability structure used in differential privacy.

Mutual information generalization bound

Assume the loss is sub-Gaussian under independent marginals:

$$\log \mathbb{E} \exp\{\lambda(\ell(W, Z) - \mathbb{E}\ell(W, Z))\} \leq \frac{R^2 \lambda^2}{2}.$$

Then an information-theoretic generalization bound is

$$|\mathbb{E}[\Delta_A(S)]| \leq \frac{1}{n} \sum_{i=1}^n \sqrt{2R^2 I(W; Z_i)}.$$

A coarser but useful version is

$$|\mathbb{E}[\Delta_A(S)]| \lesssim \sqrt{\frac{2R^2 I(W; S)}{n}}.$$

Interpretation

If the learned model contains little information about any single training point, it generalizes better.

Distributed and federated generalization

Suppose there are K nodes, each with local dataset S_k , and local output

$$W_k = A_k(S_k).$$

For local empirical-risk minimizers with convex losses, a distributed bound has the form

$$\mathbb{E}[\Delta_A(S)] \leq \frac{1}{K^2} \sum_{k=1}^K \psi^{*-1} \left(\frac{I(W_k; S_k)}{n} \right).$$

Why the factor K matters

Distributed learning can reduce generalization error because each released local model carries only partial information, and aggregation averages local instabilities.

This is the learning-theoretic version of privacy amplification by decentralization/aggregation.

Privacy constraints imply information constraints

If each local algorithm A_k is ε -locally differentially private, then the information that W_k can reveal about one local sample is limited. Under sub-Gaussian loss assumptions,

$$\mathbb{E}[\Delta_A(S)] \leq \frac{1}{K} \sqrt{\frac{2R^2 \min\{\varepsilon, (e-1)\varepsilon^2\}}{n}}.$$

Interpretation

Privacy noise is not only a security tool. It also regularizes the learning algorithm by reducing the dependence of W on individual data points.

Privacy can improve stability, but too much noise can hurt optimization accuracy.

Communication constraints also control generalization

If each node can transmit only B bits to the aggregator, then W_k can take at most 2^B possible values. Therefore

$$I(W_k; S_k) \leq H(W_k) \leq B \log 2.$$

This gives

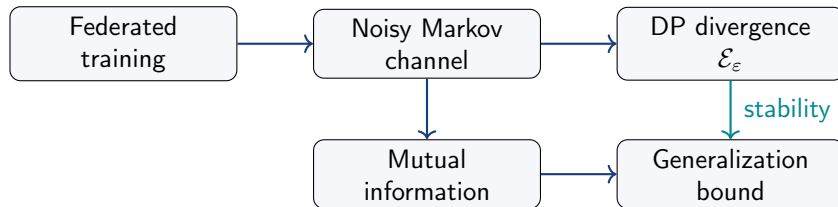
$$\mathbb{E}[\Delta_A(S)] \leq \frac{1}{K} \sqrt{\frac{2R^2(\log 2)B}{n}}.$$

Information-theoretic meaning

A communication bottleneck limits mutual information, and limited mutual information limits generalization error.

In federated systems, compression, quantization, and bit budgets are therefore statistical regularizers.

How the pieces connect



Main message

Privacy, communication, and generalization are all governed by how much information the training transcript and final model reveal about individual samples.

Relation to information theory, learning and statistics

Information theory

- Divergences measure distinguishability.
- Markov kernels contract information.
- Bit budgets bound entropy and mutual information.

Learning theory

- Generalization is stability to replacing samples.
- Mutual information bounds expected risk gaps.
- DP implies algorithmic stability.

Statistics

- Noise changes estimator bias and variance.
- Subsampling affects variance and privacy.
- Local data heterogeneity affects aggregation.

Practical design rules from the theory

Design knob	Mathematical effect	Practical consequence
Noise σ	decreases $\theta_\epsilon(\cdot)$	stronger privacy, lower leakage
Sampling rate q	multiplies leakage	smaller batches amplify privacy
Projection radius ρ	controls kernel diameter	bounds sensitivity of updates
Lipschitz constant L	controls one-step change	clipping helps privacy
Communication B	bounds $I(W; S)$	compression can improve generalization
Number of nodes K	averages local gaps	distributed learning can reduce instability

The theory explains why DP-SGD practices such as clipping, noise, subsampling, and limited communication are statistically meaningful.

Limitations and possible extensions

Limitations of the clean theory

- Convexity and smoothness are strong assumptions for modern deep networks.
- Bounds may be loose because they use worst-case sensitivity and worst-case contraction.
- IID assumptions do not fully capture non-IID federated data.

Natural extensions

- Replace global Lipschitz constants with data-dependent or local sensitivity.
- Analyze heterogeneous clients through conditional mutual information.
- Combine privacy accounting with optimization error and communication compression.

- Qinbin Li et al., *A Survey on Federated Learning Systems: Vision, Hype and Reality for Data Privacy and Protection*.
- Shahab Asodeh and Flavio P. Calmon, *Differentially Private Federated Learning: An Information-Theoretic Perspective*.
- Leighton P. Barnes, Alex Dytso, and H. Vincent Poor, *Improved Information Theoretic Generalization Bounds for Distributed and Federated Learning*.
- Cynthia Dwork et al., foundational work on differential privacy.
- McMahan et al., *Communication-Efficient Learning of Deep Networks from Decentralized Data*.

Thank you

Questions?

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